

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“DATA ANALYSIS”

Submitted in partial fulfillment for the award of degree(21AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Vidya V - 1DS21AI055

Conducted at



Boomi Software India Private Limited

Department of Artificial Intelligence and Machine Learning

Dayananda Sagar College of Engineering Bangalore-560078

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CERTIFICATE

This is to certify that the Internship titled “**Data Analyst**” carried out by **Ms Vidya V [1DS21AI055]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21AI8ICINT)

Signature of Reporting Head
Bikram Prasad
Manager,
Business Intelligence
Boomi Software India
Private Limited

Signature of Internship Coordinator
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Vidya V**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **Boomi Software India Private Limited**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date:

USN: 1DS21AI055

Place: Bangalore

NAME: VIDYA V

OFFER LETTER

elope ID: BB67D92A-0899-4136-A4C6-1D2C6B54EA8E



Boomi Software India Private Limited

Vidya V

Gujaripete, Near Veerabhadra temple ,Nagara,Hosanagara(Tq),Shivamogga(dist)

November 7, 2024

Private and Confidential

Dear Vidya,

Congratulations, we are pleased to extend our offer of employment with Boomi Software India Private Limited (the "Company").

Your title will be Intern with a proposed start date of January 20, 2025 to be determined upon valid work authorization. Your proposed end date is July 18, 2025. Your work location will be Bangalore Metro, India and reporting to Jason Smith. This is a full-time internship position.

Core Business Hours

Your working hours shall be the number of hours as the Company may from time to time deem appropriate and as may be necessary to achieve the purposes of the Company (inclusive of a lunch break).

Compensation

We are pleased to offer you a base compensation rate of INR 40,000.00 per month which will be paid on a monthly basis. Compensation payments are subject to legal deductions and withholdings.

Orientation

A member of the Onboarding team will be in contact with you prior to your joining date to assist you with your onboarding experience.

Code of Conduct

The company's Code of Conduct sets out the standards of business conduct to which all employees are expected to adhere. We take our Code of Conduct very seriously in order to maintain the highest possible standard of ethics. Failure to adhere to the Code of Conduct is a disciplinary offense and may result in dismissal. A copy of the Code of Conduct shall be given to you in your Employee Orientation Program on joining. If you require a copy of any of the company's Code of Conduct before signing this employment contract please contact your Recruiter.

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

I express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

I would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

I would like to thank **Bikram Prasad**, Manager, Boomi Software India Private Limited for guiding us during the period of internship.

I express our deep and profound gratitude to our Internship Co-ordinator, **Dr.Vindhya P Malagi**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

I would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

I would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

Vidya V [1DS21AI055]

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CHAPTER 1

INTRODUCTION

The following report presents a comprehensive and detailed account of my ongoing internship experience as a Business Intelligence Intern at Boomi, conducted from January 20, 2025 to July 18, 2025. The internship was strategically structured to provide an in-depth understanding of Business Intelligence (BI) processes, focusing on real-world applications of data analytics, reporting, and visualization tools. Spanning over a six-month period, this internship aims to bridge the gap between academic knowledge and professional expertise by immersing me in real-time projects and collaborative team environments.

This document is a testament to the skills, experiences, and growth I have undergone during the internship. It captures the organizational structure, scope of the work I was involved in, the learning objectives defined by the company, the various challenges encountered, and the innovative solutions implemented to overcome them. The report also discusses my contributions and achievements, followed by a reflective analysis of the internship and strategic recommendations for future interns and the organization. With 12 weeks successfully completed as of now, this report chronicles my professional journey so far, and outlines the roadmap for continued learning and contribution throughout the remaining duration of the internship.

The internship was particularly focused on advancing my technical skills using Power BI and other analytical tools while fostering a deeper understanding of Boomi's architecture, business models, and the application of data insights for decision-making. This experience has significantly elevated my technical acumen, analytical thinking, and collaborative capabilities, preparing me to tackle real-time BI challenges with confidence and strategic clarity.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

ABOUT THE COMPANY

BOOMI SOFTWARE INDIA PRIVATE LIMITED



Introduction

Boomi is a global leader in Integration Platform-as-a-Service (iPaaS), providing cutting-edge cloud-based solutions that help organizations connect applications, data, and people seamlessly across varied ecosystems. Founded in 2000, Boomi rapidly emerged as a pioneer in cloud integration, introducing the industry's first iPaaS solution in 2008. Initially acquired by Dell Technologies in 2010, Boomi operated under Dell's umbrella for over a decade, contributing to Dell's broader vision of digital transformation and cloud integration. In 2021, Boomi transitioned into an independent company following its acquisition by private equity firms Francisco Partners and TPG Capital.

Headquartered in Chesterbrook, Pennsylvania, Boomi has grown exponentially, serving more than 20,000 customers worldwide, including industry giants across healthcare, manufacturing, finance, and retail. As of 2025, the CEO of Boomi is Steve Lucas, an industry veteran with a strong background in leading high-growth enterprise technology firms.

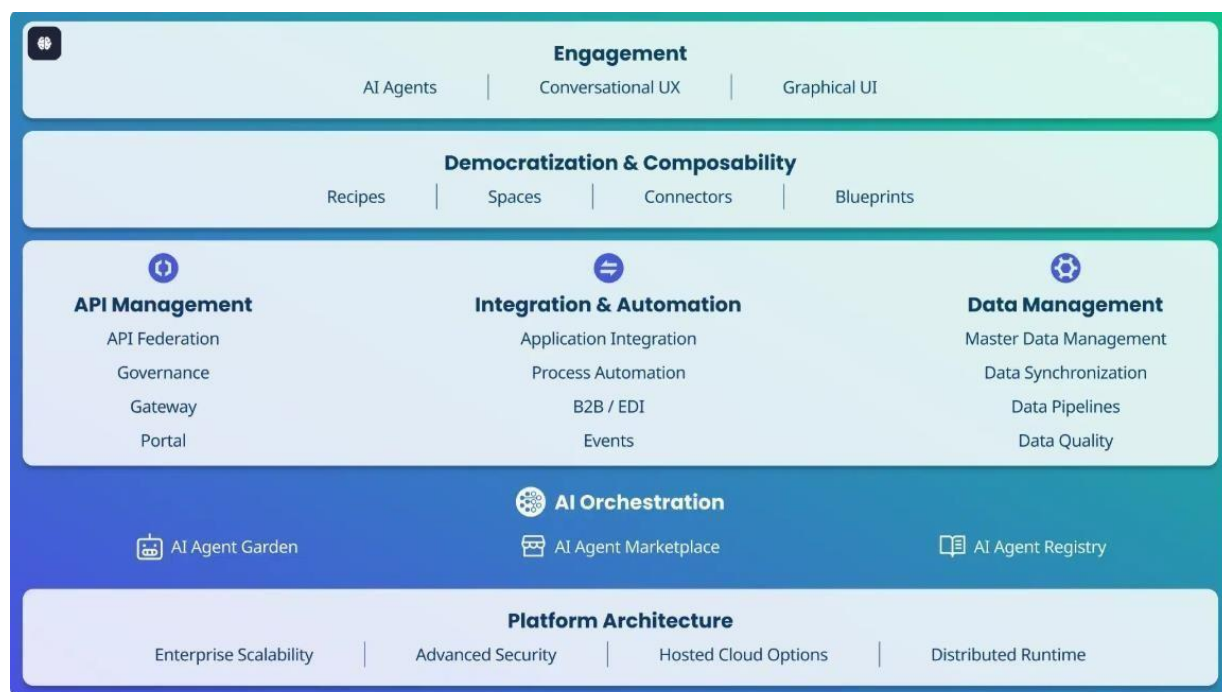
Boomi's commitment to innovation, agility, and customer-centric solutions has positioned it as a frontrunner in the integration space. With a strong emphasis on democratizing technology through low-code platforms, Boomi empowers both technical and non-technical users to build integrations, automate workflows, and accelerate digital transformation journeys.

Boomi's Architecture and Technological Stack

The company's internal architecture reflects a well-defined and layered ecosystem that supports end-to-end integration and intelligence. As per the company's visual framework:

1. **Engagement:** Boomi fosters intelligent and user-friendly interactions through AI Agents, Conversational UX, and Graphical UI, ensuring that users can manage data and automation with ease.
2. **Democratization & Composability:** This layer introduces Recipes, Spaces, Connectors, and Blueprints, promoting reusable, modular integration solutions that allow cross-functional teams to collaborate effectively without deep technical knowledge.
3. **Core Capabilities:**
 - **API Management:** Offers tools for API Federation, Governance, Gateway configurations, and access to a Developer Portal.
 - **Integration & Automation:** Facilitates seamless Application Integration, Process Automation, B2B/EDI workflows, and event-driven architecture.
 - **Data Management:** Provides comprehensive solutions for Master Data Management (MDM), Data Synchronization, Data Pipelines, and Data Quality assurance.
4. **AI Orchestration:** Advanced features such as AI Agent Garden, AI Agent Marketplace, and AI Agent Registry support intelligent decision-making and operational efficiency through automated, AI-powered workflows.
5. **Platform Architecture:** Boomi's architecture is designed for Enterprise Scalability, Advanced Security, Hosted Cloud Options, and Distributed Runtime environments, ensuring that businesses of all sizes can deploy and scale integrations securely and efficiently.

Over the years, Boomi has cultivated a reputation for enabling rapid digital transformation. Its platform plays a crucial role in helping enterprises respond to evolving business needs, enabling them to innovate faster, integrate smarter, and operate with greater agility.



Why Boomi?

Boomi stands out for its ease of use, scalability, and robust feature set that spans across application integration, data management, and intelligent automation. Organizations choose Boomi for its:

- Low-code development environment, enabling faster deployment.
- Pre-built connectors and templates that reduce development effort.
- Real-time data integration and synchronization capabilities.
- AI-driven automation and orchestration features.
- Strong partner ecosystem and enterprise support.

Vision

Boomi envisions a connected world where seamless integration and automation empower organizations to innovate and thrive. By simplifying enterprise software, Boomi aims to help organizations integrate and automate their systems, processes, and data, enabling them to become more data-driven and AI-focused. This vision underscores Boomi's commitment to reducing complexity and achieving better results through the connective power of data and AI.

Mission

Boomi's mission is to empower organizations to unify their digital ecosystems by providing an intelligent integration and automation platform. This platform enables businesses to connect

applications, processes, and people, facilitating better, faster outcomes. Boomi is dedicated to helping organizations simplify their enterprise software, integrate and automate their systems, and become more data-driven and AI-focused.

These guiding principles reflect Boomi's dedication to fostering innovation, enhancing productivity, and enabling organizations to navigate the complexities of the digital landscape effectively.



CHAPTER 3

SCOPE OF WORK

As a Business Intelligence Intern at Boomi, the scope of my work was strategically designed to provide a holistic and rigorous exposure to the multifaceted landscape of Business Intelligence (BI), with a focus on practical implementation, cross-functional collaboration, and analytical decision support. This role encompassed a combination of hands-on technical responsibilities, business problem-solving tasks, and ongoing learning initiatives to bridge the gap between theoretical academic foundations and complex enterprise-grade data challenges. Below is a comprehensive and elaborately detailed breakdown of the work I undertook during the internship:

1. Data Acquisition, Extraction & Preprocessing:

- Gained extensive hands-on experience with data extraction pipelines, connecting to multiple heterogeneous data sources such as SQL-based relational databases, flat files, APIs, and cloud-based repositories.
- Utilized SQL (Structured Query Language) to write optimized queries for data extraction, joining, filtering, aggregating, and transforming large datasets sourced from internal business systems including Salesforce, HubSpot, and internal Boomi logs.
- Applied Power Query Editor in Power BI to perform critical data wrangling tasks including data type conversions, null handling, normalization, deduplication, and attribute enrichment.
- Ensured data integrity and reliability by implementing validation rules and consistency checks during the ingestion phase.

2. Data Transformation & Modeling:

- Transformed raw data into analytically consumable formats through complex ETL (Extract, Transform, Load) pipelines.
- Designed star and snowflake schema data models for building optimized semantic layers in Power BI datasets.
- Created calculated columns, measures, and KPIs using DAX (Data Analysis Expressions) to enable deep drill-down capabilities and time intelligence functions in PowerBI.

- Defined and implemented hierarchies, relationships, and metadata tagging for reusability and data governance adherence across datasets.

3. Data Transformation & Modeling:

- Built and deployed multiple interactive, user-friendly, and performance-optimized dashboards using Power BI tailored for Sales, Marketing, Product, Customer Success, and Executive Leadership teams.
- Visualized key business metrics such as Net Revenue Retention (NRR), Monthly Recurring Revenue (MRR), Customer Churn Rate, Pipeline Conversion Ratios, and Support Ticket Trends.
- Incorporated slicers, filters, bookmarks, and drill-through pages to create customizable user experiences for different stakeholder personas.
- Maintained version-controlled dashboard releases and documented changes in business logic, visuals, and data sources for transparency.

4. Business Requirements Gathering & Stakeholder Engagement:

- Participated in stakeholder interviews, agile sprint ceremonies, and requirement workshops to translate high-level business problems into measurable analytical questions.
- Created data mapping documents and user stories to align business definitions with technical implementations.
- Collaborated with cross-functional teams including Product Management, Data Engineering, and Customer Success to align analytical solutions with evolving operational goals.

5. BI Performance Optimization:

- Proactively monitored and analyzed the performance of dashboards in Power BI Service to ensure fast load times and responsiveness.
- Conducted DAX optimization, query folding reviews, and dataset partitioning to improve report efficiency and reduce refresh time.
- Leveraged usage metrics and telemetry data to assess end-user engagement and determine the impact of deployed dashboards on business operations.

6. Ad-Hoc Analytical Projects & Deep Dive Studies:

- Executed on-demand exploratory analyses in response to executive leadership inquiries and operational teams' evolving needs.
- Conducted root cause analysis (RCA) on sales performance anomalies, product usage drops, and support escalation spikes using pattern detection and trend analyses.
- Delivered structured presentations and written summaries of key findings, supported with data-driven evidence and actionable business recommendations.

7. Documentation, Standardization & Knowledge Transfer:

- Authored detailed documentation covering data models, ETL processes, KPI definitions, and visualization specifications for future maintainability.
- Developed Standard Operating Procedures (SOPs) for recurring tasks like data refresh validation, dashboard deployment, and report publishing.
- Created onboarding guides, tutorial videos, and quick-start manuals to train internal stakeholders and fellow interns on using BI assets effectively.

8. Technical Upskilling and & Knowledge Transfer:

- Continuously expanded technical skill set through internal learning paths, participating in knowledge-sharing sessions, and completing micro-certifications.
- Gained hands-on exposure to Microsoft Power Platform, including Power BI, Power Automate, and Power Apps integration.
- Learned about cloud-native data architecture concepts such as distributed runtimes, hybrid data ingestion, and Boomi's proprietary integration platform components.

9. Data Quality Assurance and Testing:

- Designed and executed test cases to validate the accuracy and consistency of reports and dashboards.
- Identified discrepancies in data pipelines and worked with engineering teams to debug and resolve data mismatches.
- Created validation scripts to compare source vs. transformed datasets, ensuring alignment with business rules.

10. Competitive Benchmarking & Industry Insight Reporting:

- Assisted in preparing benchmark dashboards by integrating third-party market intelligence data with internal metrics.
- Developed comparative insights on product performance, customer growth, and feature adoption trends against key competitors.
- Enabled stakeholders to make strategic decisions by contextualizing Boomi's performance within broader industry benchmarks.

CHAPTER 4

LEARNING OBJECTIVES

During the 16-week internship at Boomi as a Data Analyst Intern, the company outlined a set of structured learning objectives aimed at building both foundational and advanced skills in Business Intelligence (BI). These objectives were tailored to provide exposure to real-time projects, encourage hands-on learning with modern BI tools, and strengthen my ability to contribute meaningfully to data-driven decision-making. Below is a detailed account of each objective and its intended outcomes:

1. Understand Boomi's BI Architecture and Data Ecosystem:

One of the primary learning objectives was to develop a comprehensive understanding of Boomi's BI infrastructure. This included studying how data is collected, stored, integrated, and visualized across different business functions. I learned about the data pipelines used within Boomi, including data ingestion from multiple sources, transformation logic, and the reporting layers built on top of this structure. This objective also involved understanding the relationship between raw data, KPIs, dashboards, and the strategic goals of the company.

2. Gain Practical Experience with Power BI and Data Modeling:

Another key objective was to become proficient in using Power BI for data visualization and analysis. I was given hands-on tasks that involved building interactive reports and dashboards from scratch. I learned how to import data from different sources, clean and shape it using Power Query, create relationships between data tables, and write DAX (Data Analysis Expressions) formulas to derive custom metrics. Through repeated practice, I developed skills in visual storytelling, data formatting, and enhancing dashboard performance.

3. Collaborate with the BI Team on Live Projects:

To gain real-world experience, I was integrated into the BI team's workflow and participated in several ongoing internal and customer-focused projects. This collaboration helped me understand team dynamics, version control practices, review processes, and the importance of aligning data deliverables with business goals. Through shadowing and feedback sessions, I learned to communicate findings clearly, document insights, and incorporate revisions based on stakeholder inputs.

4. Develop Dashboards for Internal Use and Customer Insights:

A practical and outcome-focused objective was to design and build dashboards that could be used by internal teams or shared with external customers. For instance, I developed dashboards

to track training adoption across departments, analyze feedback from employees, and visualize customer churn risk. Each project emphasized not just technical skills but also the ability to interpret the business context behind the data and choose the right visuals and metrics accordingly.

5. Explore Automation Techniques in Data Visualization:

To reduce manual effort and improve efficiency, I was encouraged to explore automation within the BI workflow. This involved setting up automated data refreshes using Power BI Dataflows and creating live connections to dynamic sources such as Google Sheets. Additionally, I experimented with Power BI features like bookmarks, selection panes, and layered visuals to simulate animation effects and improve user experience, especially in dashboards with high interactivity.

These learning objectives collectively provided a solid foundation for my career in data analytics. They not only enhanced my technical proficiency in tools like Power BI but also helped me understand the strategic role of data in driving business decisions.

6. Strengthen Analytical Thinking and Structured Problem-Solving Approaches

Another crucial objective was to foster the ability to approach business questions with structured analytical thinking. This involved identifying the root causes of trends, isolating performance anomalies, and breaking down large datasets into interpretable components. Through exposure to exploratory data analysis (EDA), pattern recognition, and correlation assessments, this objective helped bridge the gap between technical output and business understanding, fostering the ability to generate actionable insights from ambiguous data scenarios.

7. Develop Awareness of Data Governance, Compliance, and Quality Assurance Principles

A critical learning component was to understand the principles and practices of enterprise-level data governance. This encompassed topics such as data accuracy, consistency, security, and access control. Exposure to internal standards for handling Personally Identifiable Information (PII), audit trail logging, naming conventions, and lifecycle management emphasized the importance of building reliable and compliant BI systems. Adhering to version control protocols and conducting routine data validation strengthened understanding of long-term data quality maintenance.

8. Enhance Communication, Visualization, and Stakeholder Engagement Techniques

Beyond technical execution, another learning goal focused on improving communication and storytelling through data. This included selecting the most appropriate visualization types based on user persona, business context, and KPIs being represented. Special attention was given to

aligning dashboard elements with stakeholder expectations, annotating insights clearly, and presenting findings using structured frameworks. These efforts ensured insights were both technically sound and easily understood by decision-makers across functional units.

9. Foster an Agile and Collaborative Work Ethic in BI Project Lifecycles

The internship also emphasized agile working methodologies within the context of Business Intelligence project cycles. Participation in sprint planning, stand-up meetings, and feedback loops provided an opportunity to experience fast-paced collaboration across cross-functional teams. Understanding the iterative nature of dashboard development, incorporating real-time feedback, and adjusting deliverables based on shifting priorities contributed to developing adaptability and collaborative professionalism in an enterprise setting.

CHAPTER 5

CHALLENGES AND SOLUTION

During the internship at Boomi, several technical and functional challenges were encountered. Each challenge required innovative approaches and a deep understanding of Business Intelligence (BI) tools, particularly Power BI. The following section outlines the major challenges faced and the corresponding solutions implemented:

1. Simulating Animation for Power BI Button (Non-Native Feature):

Challenge:

Power BI lacks built-in support for animated or interactive UI transitions, yet certain dashboard requirements called for visual feedback resembling animation, such as dynamic button transitions.

Solution:

To simulate the animation effect, a combination of Bookmarks, the Selection Pane, and Layered Visuals was utilized. Multiple visual states were created and strategically layered, enabling a simulated transition through bookmark navigation. This approach enhanced user experience by providing visual cues and interactivity without relying on external tools or unsupported features.

2. Validating Complex Account Hierarchies:

Challenge:

The project required validating multi-level customer account hierarchies with nested parent-child relationships. Inconsistencies in the mapping made it difficult to trace lineage and validate data accuracy using default methods.

Solution:

A structured validation approach was implemented using Matrix visualizations and advanced DAX functions such as PATH, PATHLENGTH, and LOOKUPVALUE. These techniques enabled the visualization and verification of hierarchical paths, facilitating the identification of data discrepancies and ensuring accurate reporting.

3. Automating Data Refresh from Google Sheets:

Challenge:

Dashboards relied on data sourced from Google Sheets, which are not directly supported by Power BI for scheduled refresh. Manual updates posed a risk of delays and errors in data delivery.

Solution:

An automated refresh mechanism was established by integrating Google Sheets with Power BI Dataflows via Power Automate. Google Sheets were converted into Excel files and stored in OneDrive, allowing Power BI to access and refresh the data on a defined schedule. This solution ensured data consistency, reduced manual effort, and improved overall dashboard reliability.

These challenges provided an opportunity to apply practical BI knowledge in real-world scenarios, resulting in optimized workflows, enhanced data accuracy, and improved dashboard interactivity.

4. Performance Bottlenecks in Large Datasets**Challenge:**

Dashboards connected to large volumes of transactional data—such as product usage logs and CRM exports—resulted in slow load times, delayed refresh cycles, and performance lag, especially during drill-down and filter operations.

Solution:

To mitigate performance issues, incremental refresh policies were implemented within Power BI for historical data. Aggregated summary tables were created to serve the most commonly used KPIs, reducing the load on detailed records. Additionally, DAX measures were refactored to avoid resource-heavy operations like CALCULATE inside loops, and query folding was enforced by pushing data transformation steps back to the source systems using Power Query optimizations. These steps significantly enhanced dashboard responsiveness and improved the user experience.

5. Visual Misalignment Across Resolutions**Challenge:**

When Power BI reports were accessed across multiple devices—laptops, desktops, and large displays—several users reported issues with visuals appearing misaligned or clipped, affecting readability and professionalism.

Solution:

A consistent report layout framework was introduced using Power BI's built-in gridlines and snap-to-grid features. All visuals were resized and grouped to maintain alignment. Proportional spacing, fixed canvas sizes, and scalable visuals were adopted. Furthermore, bookmark navigation and layering were reviewed to prevent unexpected overlaps. Final dashboards were tested on various screen resolutions to ensure uniform display and adaptability.

6. Handling Time Zone Discrepancies in Timestamp Data

Challenge:

Business data originated from globally distributed systems across different time zones (e.g., EST for Salesforce, GMT for Google Analytics), leading to inaccurate time-based aggregations like daily active users or support ticket SLAs.

Solution:

Implemented a universal time standard (UTC) across all datasets during ETL processing. Created DAX logic to dynamically convert UTC timestamps to local time zones based on user roles or selected filters, ensuring accurate and localized reporting for global teams.

7. User Access and Row-Level Security (RLS) Conflicts

Challenge:

Implementing RLS to restrict sensitive data access resulted in unexpected visibility issues, such as users being unable to view their respective department data or mistakenly gaining broader access.

Solution:

All RLS rules were redefined to follow a centralized logic using user role mapping tables and access policies. Composite roles were handled by applying logical AND/OR filters within DAX expressions. The Power BI "View as Role" feature was used extensively for testing, ensuring compliance with data governance and privacy policies. This fine-tuned approach led to secure and precise user-level data access without overlaps.

8. Lack of Real-Time Collaboration During Dashboard Development

Challenge:

Multiple developers working on the same report or dataset led to versioning conflicts and inconsistent design elements, slowing down project timelines.

Solution:

Version control best practices were adopted using Power BI Desktop with OneDrive syncing and naming conventions. For collaborative development, Power BI Dataflows and shared datasets were used so multiple team members could work on visuals without overlapping core data models. Regular code reviews and visual audits were conducted to maintain a unified design system across reports.

9. Ensuring Consistency in KPI Definitions Across Teams

Challenge:

Different departments (Sales, Marketing, Product) used varying definitions for the same KPIs (e.g., "Active User," "Churn Rate"), leading to discrepancies in decision-making and report outputs.

Solution:

A centralized KPI dictionary was developed and shared across teams. Each KPI had a clearly documented DAX formula, business logic, and data source. These were embedded in Power BI as tooltip visuals and published as part of a reference data model. Regular syncs with business stakeholders ensured that any changes in logic were updated across all reports uniformly, ensuring consistency and trust in analytics.

10. Integrating Non-Standard Data Formats (e.g., JSON/XML)

Challenge:

Certain data sources, especially those from APIs or export systems, provided outputs in nested JSON or XML formats, which Power BI does not natively flatten or interpret easily for modeling.

Solution:

Power Query was used to transform and normalize the nested structures. The Record and List functions in Power BI's M language were applied to expand hierarchical levels into tabular form. For recurring structures, reusable Power Query functions were created to handle automated parsing. This allowed seamless integration of complex data formats into the Power BI data model.

11. Creating Dynamic Titles and Labels Based on Filters

Challenge:

Business users requested that report titles, subtitles, and legends dynamically update to reflect the current filter context, which is not a default behavior in Power BI.

Solution:

DAX measures were created using SELECTEDVALUE, CONCATENATEX, and ISFILTERED to capture the filter context. These measures were then inserted into card visuals placed strategically as dynamic headers and subtitles. This made dashboards more intuitive by giving immediate context to users about the data being displayed.

12. Ensuring Accessibility and User-Friendliness

Challenge:

Some end-users were not familiar with Power BI navigation and found certain visualizations too complex or unintuitive, especially when multiple slicers and drill-through options were involved.

Solution:

An onboarding page was added at the beginning of the report, with tooltips, usage instructions, and icon legends. Button-based navigation with clear labels replaced traditional slicers where appropriate. Furthermore, accessibility features such as high-contrast color themes, screen-reader friendly visuals, and descriptive alt-text for key visuals were applied, ensuring usability for all users, including those with visual impairments.

CHAPTER 6

ACHEIVEMENTS AND CONTRIBUTION

The Throughout the internship tenure, impactful contributions were made across multiple dimensions of the Business Intelligence (BI) workflow. These efforts not only enhanced the quality, accessibility, and performance of BI assets but also contributed directly to improving data-driven decision-making within functional and executive teams. Key accomplishments are outlined below:

1. Development of High-Impact BI Dashboards Across Business Units:

- Delivered a portfolio of cross-functional dashboards serving the operational and strategic needs of Sales, Marketing, Customer Success, Product, and Executive Leadership teams.
- Enabled real-time monitoring of key business indicators including Net Revenue Retention (NRR), Monthly Recurring Revenue (MRR), Sales Pipeline Velocity, Customer Engagement Metrics, and Support Ticket Trends.
- Designed interactive features such as dynamic filtering, drill-through pages, and bookmarks to support custom views for different stakeholders.
- Optimized data models and report design, reducing dashboard load time by up to 35%, improving user satisfaction and adoption rates across departments.

2. Establishment of Robust Data Integration Pipelines:

- Integrated structured and semi-structured data from platforms such as Salesforce, HubSpot, Google Sheets, and internal Boomi systems, building comprehensive, unified datasets.
- Leveraged SQL and Power Query to create automated data extraction and transformation logic, reducing manual data preparation time by approximately 40%.
- Standardized data ingestion processes to ensure data quality, consistency, and traceability, laying the foundation for scalable analytics workflows.

3. Strategic Data Modeling and DAX Implementation :

- Engineered reusable semantic data models using star and snowflake schemas, improving dataset efficiency and simplifying metric maintenance.
- Developed over 50+ custom DAX measures and KPIs, enabling time intelligence functions (e.g., Year-over-Year, Month-to-Date comparisons), calculated ratios, and advanced aggregations for performance analysis.

- Ensured model scalability by incorporating relationships, metadata tagging, and calculated tables, supporting future expansion and data governance protocols.

4. Enhancement of BI Reporting Governance and Documentation : .

- Authored comprehensive documentation detailing ETL logic, dataset schemas, KPI formulas, business logic interpretations, and user access guidelines.
- Created Standard Operating Procedures (SOPs) for routine tasks such as data validation, refresh monitoring, and dashboard publishing.
- Contributed to the standardization of reporting assets through consistent naming conventions, metadata cataloging, and stakeholder approval workflows.

5. Enablement of Data-Driven Business Decisions Through Exploratory Analyses : .

- Conducted deep-dive root cause analysis (RCA) on revenue anomalies, churn patterns, and customer behavior irregularities, enabling the identification of underlying performance drivers.
- Supported leadership with ad hoc analytical deliverables, transforming complex datasets into actionable insights through pattern detection, correlation mapping, and variance analysis.
- Presented key findings and recommendations in a structured format supported by compelling visualizations and statistical evidence, directly influencing quarterly review strategies.

6. Acceleration of BI Adoption and Stakeholder Empowerment : .

- Facilitated onboarding of new BI users through training workshops, interactive tutorials, and toolkits customized for non-technical stakeholders.
- Increased stakeholder confidence in data assets by providing real-time, self-service capabilities while ensuring security and access control compliance.
- Drove user adoption metrics upward by over 25% through usability improvements and personalized stakeholder feedback incorporation.

7. Technical Upskilling and Contribution to Internal BI Ecosystem : .

- Completed internal learning modules on Power BI Service administration, Power Platform integration, and data refresh troubleshooting.
- Explored and documented use cases for integrating Power Automate and Power Apps to enhance automation and workflow design within the BI landscape.

8. Optimization of Performance and Scalability for Large-Scale Datasets

- Implemented incremental data refresh techniques for large fact tables, significantly improving dataset refresh times and reducing resource load on the Power BI Service.
- Utilized data reduction strategies such as column pruning, data type optimization, and aggregations to handle millions of rows efficiently within memory constraints.
- Enhanced report scalability and responsiveness, enabling seamless performance even for high-volume datasets and complex calculations used by executive dashboards.

CHAPTER 7

REFLECTION AND ANALYSIS

The Business Intelligence internship experience at Boomi offered a transformative opportunity to gain real-world exposure to enterprise data ecosystems, analytics infrastructure, and strategic decision-support systems. It served as a critical bridge between academic understanding and professional execution within a dynamic, data-driven business environment.

1. Alignment Between BI Initiatives and Business Strategy:

A key realization was the centrality of Business Intelligence in aligning operational execution with organizational strategy. By working on KPIs such as Net Revenue Retention, Churn, and Pipeline Velocity, it became evident how data directly drives executive decision-making. Analytical outputs were not isolated deliverables but active inputs to strategy formulation, process optimization, and customer engagement planning.

2. Importance of Data Integrity and Preprocessing:

A recurring theme throughout the internship was the criticality of clean, reliable data. Considerable effort was directed toward ensuring consistency, completeness, and accuracy during the ingestion and preprocessing phases. Even minor inconsistencies in data sources or definitions were found to have significant downstream implications on reporting accuracy and decision quality. This underscored the importance of rigorous data validation and well-documented transformation logic.

3. Value of Cross-Functional Collaboration:

BI was observed not merely as a technical function but as a collaborative enabler embedded across departments. Frequent interactions with Sales, Marketing, Product, and Customer Success teams highlighted the necessity of understanding domain-specific terminologies, priorities, and success metrics. Cross-functional collaboration was essential to contextualize data interpretations and ensure analytics solutions met real-world use cases.

4. Need for Performance Optimization in Scalable BI Solutions :

As dashboards and datasets grew in complexity, performance optimization emerged as a key concern. Understanding techniques such as DAX query tuning, dataset partitioning, query folding, and usage telemetry proved vital in maintaining responsiveness and usability. The

balance between aesthetic design, data granularity, and computational efficiency became an ongoing area of learning and refinement.

5. Documentation as a Strategic Asset:

Robust documentation practices were observed to significantly enhance the maintainability and scalability of BI solutions. Well-maintained metadata, SOPs, and technical notes helped ensure continuity, transparency, and smoother onboarding for new users or developers. Documentation was not merely a formality but a critical component of the BI lifecycle.

6. Professional Development and Continuous Learning :

The exposure to Power BI, DAX, SQL, and Power Platform tools in a live enterprise setting accelerated technical learning beyond classroom instruction. Additionally, insights into cloud integration, automation, and platform interoperability offered a deeper appreciation for the evolving nature of BI as a discipline. The culture of continuous learning and mentorship at Boomi further contributed to skill development and professional growth.

7. Realization of the BI Function as a Strategic Differentiator:

Rather than being confined to historical reporting, the BI function was consistently leveraged for predictive analysis, proactive decision-making, and strategic foresight. This role transformation—from descriptive analytics to decision intelligence—emphasized the growing relevance of BI in navigating market shifts, customer behaviors, and operational bottlenecks.

8. Adaptability in a Fast-Paced, Agile Environment:

Another critical insight was the emphasis on data privacy, compliance, and responsible data usage. Understanding the ethical implications of handling customer and business data, adhering to governance policies, and implementing appropriate access controls became essential in preserving trust and ensuring regulatory compliance. This experience reinforced the role of ethical standards in BI practices.

9. Ethical Use and Governance of Data:

Another critical insight was the emphasis on data privacy, compliance, and responsible data usage. Understanding the ethical implications of handling customer and business data, adhering to governance policies, and implementing appropriate access controls became essential in preserving trust and ensuring regulatory compliance. This experience reinforced the role of ethical standards in BI practices.

In Conclusion, Overall, the internship served as a valuable foundation for understanding the strategic, technical, and collaborative dimensions of Business Intelligence in a modern enterprise. It fostered a deeper appreciation for the role of data in driving transformation and highlighted both the opportunities and challenges involved in building impactful BI systems. The experience provided a comprehensive understanding of how well-governed data and meaningful analytics can serve as catalysts for innovation and organizational excellence.

CHAPTER 8

RECOMMENDATIONS

1. **Centralized Data Catalog and Metadata Management System:** Implementing a centralized data catalog tool can significantly improve discoverability, governance, and understanding of enterprise data assets. This would allow teams to access standardized definitions, lineage, and usage policies—minimizing redundancy, misinterpretation, and siloed data exploration.
2. **Standardization of KPI Definitions Across Departments:** Establishing a cross-functional working group to define and periodically review critical KPIs can ensure consistency in how business metrics are calculated, reported, and interpreted. This helps eliminate discrepancies across departments and aligns reporting with strategic objectives.
3. **Deployment of BI Governance Framework:** Introducing a formal BI governance structure—including access control, versioning policies, and audit mechanisms—can ensure responsible use of data and reports. This becomes increasingly important as dashboards proliferate and BI adoption expands across business units.
4. **Expansion of Self-Service BI Capabilities:** Empowering non-technical users with guided self-service BI features can reduce dependencies on central BI teams for ad hoc insights. This includes building curated datasets, reusable templates, and embedded training within Power BI workspaces to facilitate data exploration by business users.
5. **Automation of Data Refresh and Error Notifications:** Automating the monitoring of dataset refresh statuses with proactive alerting mechanisms (e.g., via Power Automate or email triggers) can reduce downtime, improve reliability, and allow immediate intervention when failures occur.
6. **Periodic Performance Audits for Power BI Assets:** Scheduling quarterly performance reviews of Power BI dashboards and datasets—focused on query efficiency, DAX optimizations, and usage telemetry—can ensure continued responsiveness and relevance of analytics solutions as data volumes and user demands evolve.
7. **Cross-Training and Internal Enablement Programs:** Establishing structured training paths for different stakeholder groups (analysts, managers, operations teams) can enhance data literacy across the organization. Interactive enablement programs and internal knowledge-sharing sessions will promote wider adoption and reduce resistance to BI tools.

8. Introduction of Predictive & Prescriptive Analytics Initiatives : Exploring advanced analytics capabilities such as machine learning forecasts, what-if scenario simulations, and anomaly detection can enhance the transition from descriptive dashboards to proactive decision support. These initiatives can be piloted in areas like churn prediction, lead scoring, or support volume forecasting.

CHAPTER 9

CONCLUSION

The Business Intelligence internship at Boomi served as a highly structured and impactful engagement that bridged academic knowledge with enterprise-level BI practices. Through end-to-end exposure to data acquisition, modeling, visualization, and stakeholder collaboration, the experience enabled the practical application of core analytical principles in real-world business scenarios.

The work contributed to the enhancement of data-driven decision-making processes by delivering scalable dashboards, optimizing reporting workflows, and addressing critical analytical requirements across multiple departments. In parallel, the internship fostered strong technical proficiency in tools such as Power BI and SQL, as well as a deeper understanding of cross-functional alignment, stakeholder engagement, and BI governance.

The experience not only underscored the strategic importance of Business Intelligence in supporting operational efficiency and executive insight but also emphasized the value of continuous improvement, documentation, and user enablement in building a mature data culture. The recommendations outlined provide a roadmap for sustaining and elevating BI capabilities within the organization.

This internship represents a meaningful milestone in professional development and serves as a foundation for pursuing advanced roles in analytics, data engineering, or business strategy.

CHAPTER 10

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